Better product decisions with Nexus

Know what changes. See what needs review. Reuse what the business trusts.

For product leaders, engineers, manufacturing planners and quality teams

A product change rarely belongs to one team. It may involve an assembly revision, a requirement, a manufacturing activity and inspection evidence. Nexus helps teams bring the relevant records together, understand their meaning and prepare a traceable basis for review.

Why the business needs it

Today, people often spend the first part of a review finding information rather than evaluating it. Different names, unclear revisions and missing associations create repeated work. A familiar part name is not enough to judge reuse, and an AI answer is not useful unless the team can inspect its evidence.

Business question	How Nexus helps the team
What needs review if this part changes?	Investigate recorded assembly, requirement and process dependencies for the selected item.
Could an existing part meet this need?	Compare classification, property values, declared units and source context.
Where is the evidence incomplete?	Expose missing associations and unanswered questions for the responsible specialists.
Which analysis can another team reuse?	Prepare records, review the intended use and publish an owned data product.

A practical engineering change example

A bearing design changes within a motor assembly. The engineer selects the intended revision, inspects recorded parent assemblies, follows requirement and process references, and reviews the supporting evidence. The team decides what needs further investigation and records the permitted action. Missing links remain questions; they do not prove that there is no impact.

Value to measure

Measure the time to assemble an impact review, locate authoritative records and compare reusable designs. Compare the pilot with the current process; do not assume a percentage saving or a guaranteed return.

A common product story across disciplines

Product, systems engineering, manufacturing and quality teams each see part of the same story. Nexus connects supplied business records through reviewed definitions and relationships while retaining their source identity.

Business view	What the team can review
Product and portfolio	Products, families, assemblies and the part revisions within the selected scope.
Systems engineering / MBSE	Available requirement, model and product relationships. Traceability is distinct from satisfaction evidence.
Manufacturing	Recorded parent/child part relationships and process or activity assignments.
Quality	Supplied specification, characteristic or inspection evidence with established ownership and source associations.

Industry ontology, in business terms

An ontology defines concepts and how they relate. A physical part differs from its design. A requirement differs from evidence that it was satisfied. The embedded industrial reference and application business definitions help teams preserve these distinctions. Customer vocabulary is aligned through reviewed mappings rather than treating similar names as equivalent.

Who does what

Role	Contribution to the business workflow
Business owner and specialists	Set the intended use, review fitness for use and authorize the release scope.
Governor agent	Coordinates mapping or merge proposals and the authorized publication workflow.
Steward agent	Assesses meaning, ambiguity and retained validation evidence; explains accepted and held findings.
Workers	Carry out authorized processing and retain results while people focus on review.

The public industrial ontology snapshot supports shared meaning; it does not establish ISO certification or approved alignment to a customer domain.

Industry scenarios: the question comes first

These are applicability scenarios. Each needs the customer's records, domain definitions, reviewed mappings and acceptance criteria. They are not claims of preconfigured sector solutions.

Industry	Business decision and functional journey
HiTech	A component revision changes. Select the intended component, trace recorded product and requirement references, compare candidates and prepare an engineering review. Measure search and comparison effort.
Oil and gas	An equipment specification changes. Bring supplied equipment, engineering document and activity records into a shared context. Help specialists identify related evidence and unresolved associations.
Food and beverages	A product specification changes. Establish reviewed product, ingredient, process and quality references within the agreed scope. Help development and quality teams inspect the evidence chain.
Pharma and life sciences	A material or specification is reviewed. Organize supplied material, process and test evidence by version, ownership and recorded relationships for accountable specialist review.

The business journey

1. Choose the product or operational question and the responsible reviewer.
2. Bring the relevant records together and confirm terms, revisions and scope.
3. Investigate recorded relationships and make missing evidence visible.
4. Review the proposal, authorize the intended action and verify its outcome.
5. Share the released result with ownership and source references.

Keep the decision boundary clear

Similarity is not interchangeability. Recorded traceability is not requirement satisfaction. Observed links do not certify a complete BOM or manufacturing readiness. Nexus supports evidence review; it does not by itself certify food safety, regulatory compliance, GxP validation or batch release.

From useful answers to reusable business knowledge

Conversation with the right context

A user asks about a selected product and follows up about its requirements or properties. Nexus keeps relevant conversation context within the authorized business scope, retrieves fresh evidence and presents business names, descriptions, property values and source references. Approved, revision-scoped mapping lessons can support later reviews as advisory guidance. Live language generation requires an available, configured model; retained memory is not model training.

Business analytics and data products

A portfolio team chooses registered business records, prepares a retained analysis snapshot and selects reporting dimensions. The team agrees measure definitions, denominators, units, revisions and effectivity before interpreting results. Reviewed tables and results can then be published as an owned product for discovery and reuse. Supported reporting models still require business review; there is no universal automatic KPI library.

Know what was approved, executed and released

Change Control brings together status and links to the responsible review workflow. A recommendation is a proposal, approval grants authority for a scope, and publication evidence confirms the retained outcome. These states remain distinct; central visibility does not replace a full enterprise change-request/change-order process.

Keep a baseline when information changes

Retained source versions and declared references show what supported a result. Supported Tracked Resource Set (TRS) notifications and source reconciliation can identify refresh needs. A change prompts review rather than silently authorizing another release. This is scoped baseline support, not global enterprise configuration management.

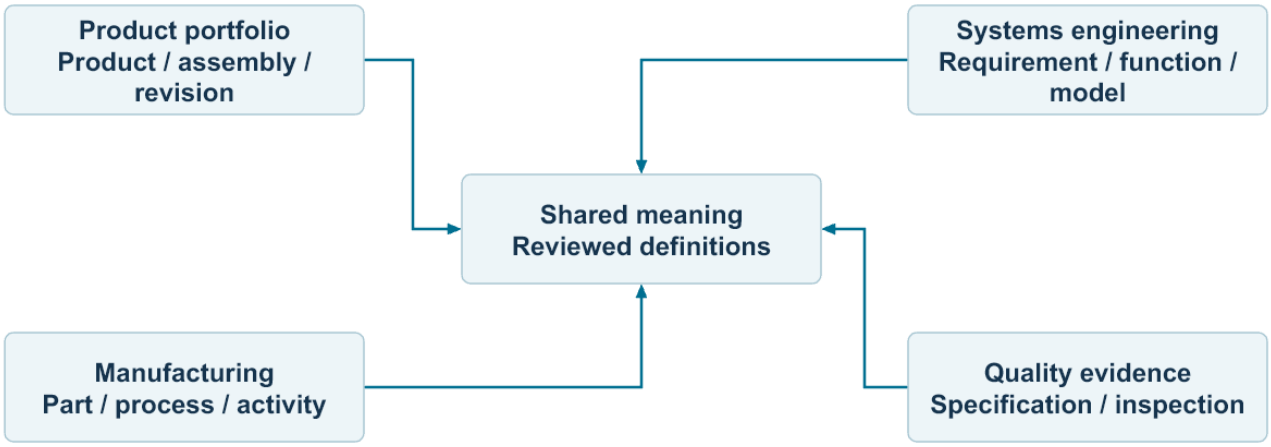
Start with one decision worth improving

Pilot focus	Agree how success will be assessed
Change review	Time to assemble the review and resolve missing associations.
Reuse and reporting	Effort to compare candidates and prepare a reusable analysis.
Accountability	Availability of owner, source/version, review outcome and publication evidence.

Demonstrate one real product question, one governed review and one reusable analysis. Expand to another domain after the business accepts the first scenario.

Connected business views and responsibilities

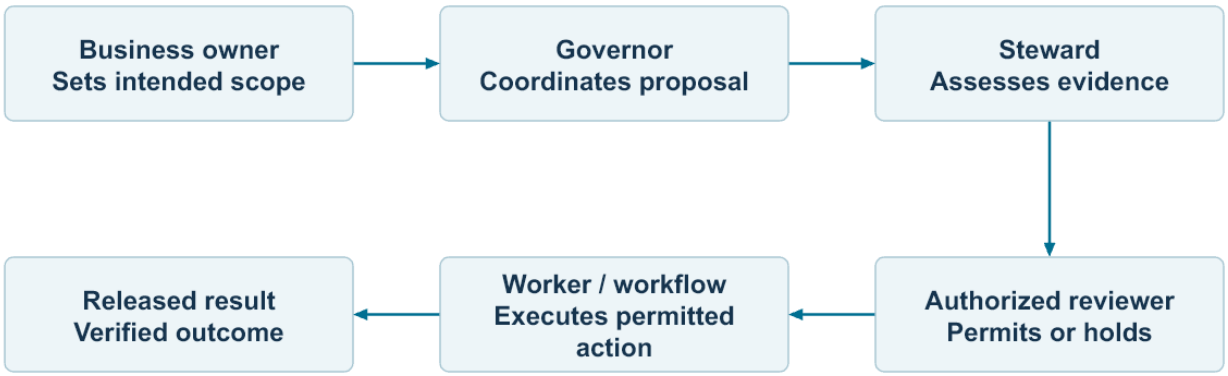
One shared meaning across product domains



Conceptual view: relationships depend on supplied records and reviewed mappings.

Conceptual view: connections depend on supplied business records and reviewed mappings.

People, agents and workers in the review



Held proposals stay unpublished. Agents and workers do not grant themselves business authority.

Business authority remains with the authorized reviewer. A held proposal remains unpublished.

Context, memory and controlled release

Remember the right context; retrieve fresh evidence



Conversation history and approved mapping lessons are separate; neither is model training.

Current conversation and approved mapping lessons serve different purposes. Neither is automatic model training.

From proposed change to a verified outcome



Held proposals stay unpublished. Approval, execution and availability remain distinct.

Review, authorization, execution and publication are distinct states. Inspect the retained outcome before use.